Especially intriguing is the appearance of the imploding star as observed from a *static, external reference frame,* that is, as seen by observers outside the star who remain always at the same fixed circumference instead of riding inward with the star's imploding matter. The star, as seen in a static, external frame, begins its implosion in just the way one would expect. Like a rock dropped from a rooftop, the star's surface falls downward (shrinks inward) slowly at first, then more and more rapidly. Had Newton's laws of gravity been correct, this acceleration of the implosion would continue inexorably until the star, lacking any internal pressure, is crushed to a point at high speed. Not so according to Oppenheimer and Snyder's relativistic formulas. Instead, as the star nears its critical circumference, its shrinkage slows to a crawl. The smaller the star gets, the more slowly it implodes, until it becomes *frozen* precisely at the critical circumference. No matter how long a time one waits, if one is at rest outside the star (that is, at rest in the static, external reference frame), one will never be able to see the star implode through the critical circumference. That is the unequivocal message of Oppenheimer and Snyder's formulas.

Is this freezing of the implosion caused by some unexpected, general relativistic force inside the star? No, not at all, Oppenheimer and Snyder realized. Rather, it is caused by gravitational time dilation (the slowing of the flow of time) near the critical circumference. Time on the imploding star's surface, as seen by static external observers, must flow more and more slowly when the star approaches the critical circumference, and correspondingly everything occurring on or inside the star including its implosion must appear to go into slow motion and then gradually freeze.

As peculiar as this might seem, even more peculiar was another prediction made by Oppenheimer and Snyder's formulas: Although, as seen by static external observers, the implosion freezes at the critical circumference, it *does not freeze at all* as viewed by observers riding inward on the star's surface. If the star weighs a few solar masses and begins about the size of the Sun, then as observed from its own surface, it implodes to the critical circumference in about an hour's time, and then keeps right on imploding past criticality and on in to smaller circumferences.